

Assignment-based Subjective Questions

1. From your analysis of the categorical variables from the dataset, what could you infer about their effect on the dependent variable?

Ans: There were 6 categorical variables in the dataset. I used Box plot to study their effect on the dependent variable ('cnt'). The inference that I could derive were: - season: Almost 32% of the bike booking were happening in season3 with a median of over 5000 booking (for the period of 2 years). This was followed by season2 & season4 with 27% & 25% of total booking. This indicates, season can be a good predictor for the dependent variable. - mnth: Almost 10% of the bike booking were happening in the months 5,6,7,8 & 9 with a median of over 4000 booking per month. This indicates, mnth has some trend for bookings and can be a good predictor for the dependent variable. - weathersit: Almost 67% of the bike booking were happening during 'weathersit1' with a median of close to 5000 booking (for the period of 2 years). This was followed by weathersit2 with 30% of total booking. This indicates, weathersit does show some trend towards the bike bookings can be a good predictor for the dependent variable. - holiday: Almost 97.6% of the bike booking were happening when it is not a holiday which means this data is clearly biased. This indicates, holiday can not be a good predictor for the dependent variable. - weekday: weekday variable shows very close trend (between 13.5%-14.8% of total booking on all days of the week) having their independent medians between 4000 to 5000 bookings. This variable can have some or no influence towards the predictor. I will let the model decide if this needs to be added or not. - workingday: Almost 69% of the bike booking were happening in 'workingday' with a median of close to 5000 booking (for the period of 2 years). This indicates, workingday can be a good predictor for the dependent variable.

2. Why is it important to use drop_first=True during dummy variable creation?

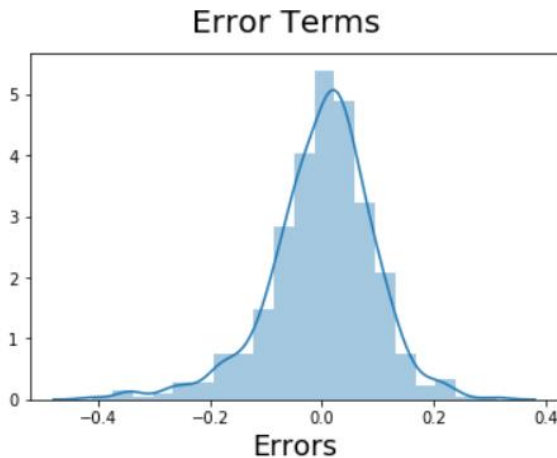
Ans: drop_first=True is important to use, as it helps in reducing the extra column created during dummy variable creation. Hence it reduces the correlations created among dummy variables.

3. Looking at the pair-plot among the numerical variables, which one has the highest correlation with the target variable?

Ans: Using the pair plot, we could see there is a linear relation between temp variable with the predictor 'cnt' and has the highest correlation with the target variable .

4. How did you validate the assumptions of Linear Regression after building the model on the training set?

Ans: The Residuals are normally distributed. Hence our assumption for Linear Regression is valid.



5. Based on the final model, which are the top 3 features contributing significantly towards explaining the demand of the shared bikes?

Ans: As per our final Model, the top 3 predictor variables that influences the bike booking are: --- -

Temperature (temp) - A coefficient value of '0.5636' indicated that a unit increase in temp variable increases the bike hire numbers by 0.5636 units. - Weather Situation 3 (weathersit_3) - A coefficient value of '-0.3070' indicated that, w.r.t Weathersit1, a unit increase in Weathersit3 variable decreases the bike hire numbers by 0.3070 units. -

Year (yr) - A coefficient value of '0.2308' indicated that a unit increase in yr variable increases the bike hire numbers by 0.2308 units. --- SO IT IS RECOMMENDED TO GIVE THESE VARIABLES UTMOST IMPORTANCE WHILE PLANNING, TO ACHIEVE MAXIMUM BOOKING. ---

The next best features that can also be considered are - - season_4: - A coefficient value of '0.128744' indicated that w.r.t season_1, a unit increase in season_4 variable increases the bike hire numbers by 0.128744 units. - windspeed: - A coefficient value of '-0.155191' indicated that, a unit increase in windspeed variable decreases the bike hire numbers by 0.155191 units.

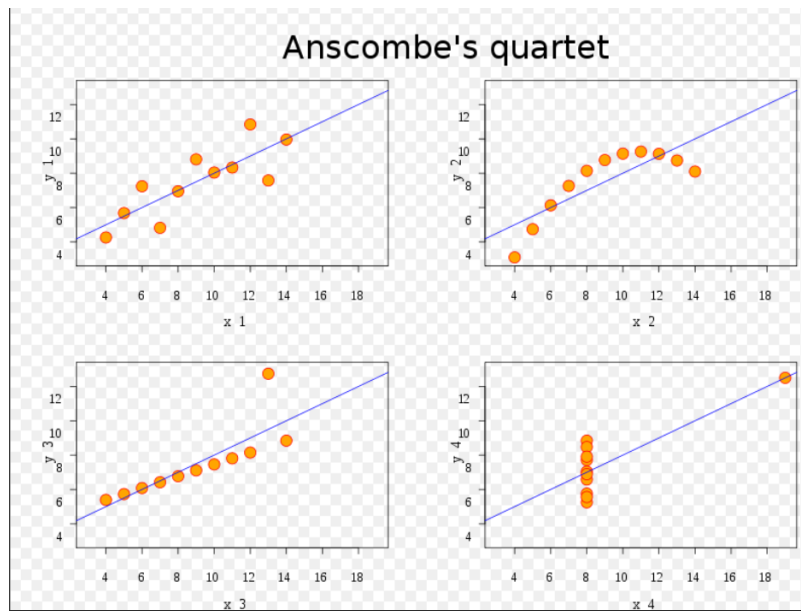
General Subjective Questions

1. Explain the linear regression algorithm in detail.

Ans: Linear regression algorithm shows a linear relationship between a dependent (y) and one or more independent (x) variables, hence called as linear regression. Since linear regression shows the linear relationship, which means it finds how the value of the dependent variable is changing according to the value of the independent variable.

2. Explain the Anscombe's quartet in detail.

Ans: Anscombe's quartet comprises four data sets that have nearly identical simple descriptive statistics, yet have very different distributions and appear very different when graphed. Each dataset consists of eleven (x, y) points.

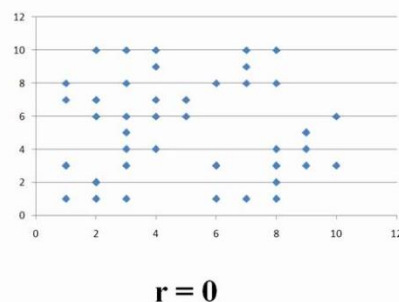


3. What is Pearson's R?

Ans: The Pearson correlation coefficient, also called Pearson's R, is a statistical calculation of the strength of two variables' relationships. In other words, it's a measurement of how dependent two variables are on one another.

Pearson's r measures the strength of the linear relationship between two variables.

Pearson's r is always between -1 and 1.



4. What is scaling? Why is scaling performed? What is the difference between normalized scaling and standardized scaling?

Ans: Scaling means that you transform your data to fit into a specific scale, like 0-100 or 0-1.

You want to scale the data when you use methods based on measurements of the distance between data points, such as supporting vector machines and the k nearest neighbors.

Scaling vs Normalization. Feature scaling (also known as data normalization) is the method used to standardize the range of features of data. Since, the range of values of data may vary widely, it becomes a necessary step in data preprocessing while using machine learning algorithms.

5. You might have observed that sometimes the value of VIF is infinite. Why does this happen?

Ans: An infinite VIF value indicates that the corresponding variable may be expressed exactly by a linear combination of other variables.

6. What is a Q-Q plot? Explain the use and importance of a Q-Q plot in linear regression.

Ans: The Q-Q plot or quantile-quantile plot is a graphical technique for determining if two data sets come from populations with a common distribution. A Q-Q plot is a scatterplot created by plotting two sets of quantiles against one another.